Learning the Importance of Authentication Factors

Natrona County School District #1 Draft Computer Science Curriculum

Benchmark	Components for Grade 1 Outcome 3		
Domain: Networks and the Internet (NI) / Standard: Cybersecurity (C)			
2.NI.C.01 **Priority**	2.NI.C.01 Notes	Step 1 is taught in Kindergarten. Step 2 and the benchmark are taught in Grade 1. Practice 7.3 Communicating About Computing	Suggested Curriculum Integration
	Step 2	Recognize cybersecurity protects one's device or information from unwanted use by others (e.g., authentication factors)	Computer Science
	Benchmark	Explain what authentication factors (e.g., login) are, why we use them, and apply authentication to protect devices and information (personal and private) from unauthorized access. Example(s) and/or Clarification: • Compare passwords to the combination on a bicycle lock. • Passwords include letters and numbers to make it more difficult to steal.	
		Learning Targets: • Explain what authentication factors are. (DOK 1 Recall) • Explain why we use authentication factors. (DOK 1 Recall) • Apply authentication factors by logging into a device. (DOK 2 Skill/Concept) • Apply authentication factors by logging into a program. (DOK 2 Skill/Concept) Vocabulary: • authentication factor (username and password that gives you authorization)	

Materials:
Locked Mystery Box, IPads, Login Sheets

Pre/Post Test Questions:
What are authentication factors?
Why do we use authentication factors?
How do we keep our authentication factors safe?

Hook:
Students will be presented with a locked mystery box, found somewhere in the building. They will be asked to make predictions as to what may be in the box. They will then be asked why the box may be locked, and how they may go about getting into the box. They will then be asked if it would be right or fair to enter the box without permission from the box's owner.

Set the mystery box aside and tell students we will circle back to the box in a bit.

At this time, introduce the ipads to students. Give a brief walkthrough of expectations for ipad care and use. Where it will be stored in the classroom, how it will travel from school and home, that it must be charged nightly, and how we will use it throughout the school day.

Charging cords will be distributed to be placed in backpacks.

Ipads will be distributed and placed in student mailboxes.

Talk to students about how many applications can help us learn, and it is important to do our best work and to only work within our own accounts. Revisit the Mystery Box briefly; if that was your box, would you like someone else to break into it? The same can be said for our accounts on our ipads.

Introduce the term, “Authentication Factors,” including login and password. Explain to students they will have a login and password that follows them through the grades for each application we use on our devices.

Explain the importance of keeping their logins and passwords private, and the importance of never ever logging into someone else’s account, much like the mystery box from before.

Revisit the mystery box at this time, share that it does in fact belong to our classroom, and that we have the key.

Unlock the mystery box, and remove the login and password sheets. Introduce them to students, showing them how to locate and use the first set of authentication factors.

At this time, students should work through how to find the app and log in.

Give students time to interact with the application before walking them through the log out process, and putting ipads and login sheets away safely.

Repeat the process of logging into different applications over the course of the next several days, troubleshooting with students that have issues.